

Ehsan Ghoreishi

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EDUCATION

Virginia Tech

Ph.D. in Computer Engineering

Blacksburg, VA, U.S.

2024-present

Virginia Tech

M.Sc. in Computer Engineering

Blacksburg, VA, U.S.

2024

Isfahan University of Technology

B.Sc. in Electrical Engineering

Isfahan, Iran

2021

RESEARCH INTERESTS

Wireless Networks, Machine Learning, Reinforcement Learning, Real-Time Systems, Optimization

RESEARCH EXPERIENCE

Design an intelligent NextG scheduler for URLLC/eMBB multiplexing

2022–present

Complex Network & Security Research Lab, Virginia Tech

- **Cyrus**: Designed an O-RAN-compliant DRL-based puncturing scheduler for URLLC/eMBB multiplexing.
- **Cyrus+**: Advanced DRL-based puncturing scheduler achieving real-time operation ($\sim 125 \mu\text{s}$) to maximize eMBB goodput while ensuring URLLC latency guarantees.
- **PuRe**: Designed a learning-based framework that (a) predicts URLLC packet arrival patterns using an LSTM model, and (b) applies DRL-based puncturing on eMBB transmissions to maximize goodput while ensuring URLLC latency guarantees.
- Conducted a survey on how machine learning techniques can enhance eMBB/URLLC multiplexing.

Satellite Communication

2025–present

Complex Network & Security Research Lab, Virginia Tech

- Investigated LSTM-driven CQI prediction methods to enable adaptive resource block allocation under long LEO propagation delays.
- Addressed the challenge of short satellite dwell times ($\sim 15 \text{ s}$ in Starlink) by exploring lightweight deep learning models for real-time training and inference.

Complex-Valued Neural Network for Wireless Modulation Classification

2020–2021

Signal Processing Research Center, Isfahan University of Technology

- Designed and implemented a complex-valued CNN model for automatic wireless modulation classification.
- Leveraged in-phase/quadrature (I/Q) signal representations to improve feature extraction and classification accuracy.

Designed and Analyzed Nonlinear Effects in Remote Wire Monitoring

2020–2021

Optics Research Center, Isfahan University of Technology

- Developed optical sensing techniques to remotely monitor structural wires under nonlinear effects.
- Modeled and analyzed nonlinear optical responses to enhance detection sensitivity and reliability.

PUBLICATIONS

Patent

Ehsan Ghoreishi, Bahman Abolhassani, Y. Thomas Hou, “A Method for Multiplexing Ultra Low Latency Data with Other Data over the Same Radio Frame,” U.S. Patent Application, filed October 2025, pending.

Papers

Ehsan Ghoreishi, Bahman Abolhassani, Wenjing Lou, Y. Thomas Hou, “RaP: Learning-based Joint Reservation and Puncturing for Efficient URLLC/eMBB Multiplexing,” in *Proc. IEEE International Conference on Computer Communications (INFOCOM)*, 10 pages, Tokyo, Japan, 2026.

Ehsan Ghoreishi, Bahman Abolhassani, Yan Huang, Shiva Acharya, Wenjing Lou, Y. Thomas Hou, “Cyrus+: A Real-Time DRL-based Puncturing Solution to URLLC/eMBB Multiplexing in O-RAN,” *IEEE Transactions on Machine Learning in Communications and Networking*.

Ehsan Ghoreishi, Bahman Abolhassani, Yan Huang, Shiva Acharya, Wenjing Lou, Y. Thomas Hou, “Cyrus: A DRL-based Puncturing Solution to URLLC/eMBB Multiplexing in O-RAN,” in *Proc. International Conference on Computer Communications and Networks (ICCCN)*, 9 pages, Hawaii, USA, 2024. (Invited Paper)

Ehsan Ghoreishi, Shiva Acharya, Wenjing Lou, Y. Thomas Hou "Enhancing 5G Efficiency: How Machine Learning is Revolutionizing eMBB and URLLC Puncturing." (To be submitted to *IEEE Wireless Communications*)

Poster.....

Ehsan Ghoreishi, Bahman Abolhassani, Yan Huang, Shiva Acharya, Wenjing Lou, Y. Thomas Hou "Cyrus: A DRL-based Puncturing Solution to URLLC/eMBB Multiplexing in O-RAN," *Office of Naval Research (ONR) MURI Meeting*, Washington D.C., USA, 2024.

AWARDS AND ACHIEVEMENTS

CCI SWVA Cyber Innovation Scholar 2025
Recipient, Commonwealth Cyber Initiative – SWVA

TEACHING EXPERIENCE

The Graduate Teaching Assistant in Introduction to Computer Networking 2022
Bradley Dept. of Electrical and Computer Engineering, Virginia Tech

The Teaching Assistant in Principles of Electronics 2021
Electrical and Computer Engineering Dept., Isfahan University of Technology

The Teaching Assistant in Electric Circuits II 2021
Electrical and Computer Engineering Dept., Isfahan University of Technology

Course Instructor in MATLAB 2022
Electrical and Computer Engineering Dept., Isfahan University of Technology

SKILLS

Programming Languages: Python, MATLAB, C/C++, Java
Machine Learning & Data Science: PyTorch, TF-Agents, TensorFlow
Engineering Tools: MATLAB, Simulink

PROFESSIONAL MEMBERSHIPS

Institute of Electrical and Electronics Engineers (IEEE) 2022-present
IEEE Communications Society (ComSoc) 2022-present